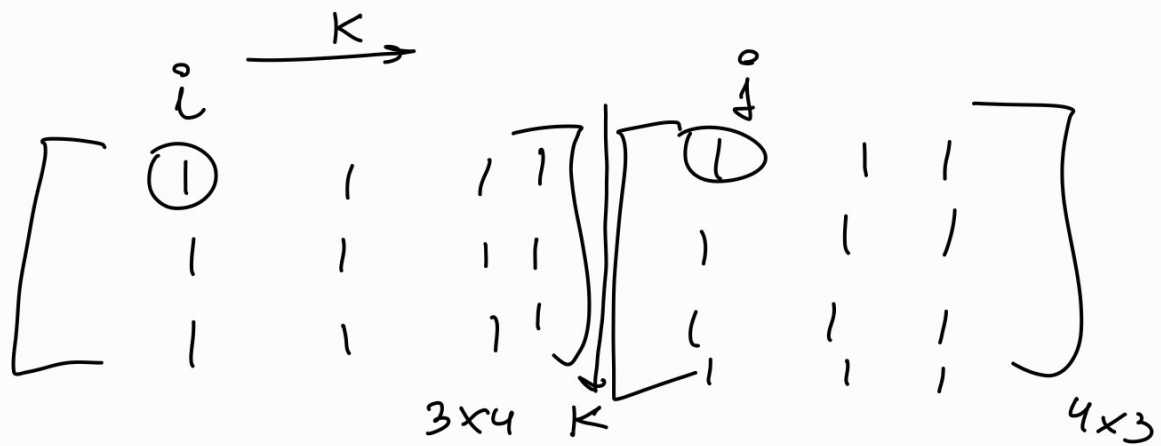




$$arr[i][K] * arr[K][j]$$

$$\begin{bmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \\ 4 & 4 & 4 \end{bmatrix}$$

$$\begin{matrix} i=0, j=0 \\ \textcircled{1} \\ 1 \end{matrix} \quad \begin{matrix} i=0, j=1 \\ \textcircled{2} \end{matrix} \quad \begin{matrix} i=0, j=2 \\ \textcircled{3} \end{matrix} \quad \begin{matrix} 2 \\ 3 \end{matrix} \quad \begin{matrix} 2 \times 3 \\ = \end{matrix} \quad \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\frac{1}{1+e^{-z}}$$

$$\frac{d}{dz} \left(\frac{1}{1+e^{-z}} \right) = -$$

$$\frac{d}{dz} \left(\frac{a}{b} \right) = \frac{\frac{d}{dz}(a) b - \frac{d}{dz}(b) a}{b^2}$$

$$- \frac{\frac{d}{dz}(1+e^{-z})}{(1+e^{-z})^2}$$

$$1 - \frac{1}{1+e^{-z}} \quad \frac{e^{-z}}{(1+e^{-z})^2}$$

$$\frac{1+e^{-z} \rightarrow e^{-z}}{1+e^{-z}} \times \frac{1}{1+e^{-z}} = \frac{\sigma(z)(1-\sigma(z))}{\sigma(z)(1-\sigma(z))}$$

$$f(x) = \begin{cases} x & x \geq 0 \\ 0 & x \leq 0 \end{cases}$$

$$MSE = \frac{1}{2n} \|\hat{y} - y\|^2$$

$$= \frac{1}{2n} \|X^T w - y\|^2$$

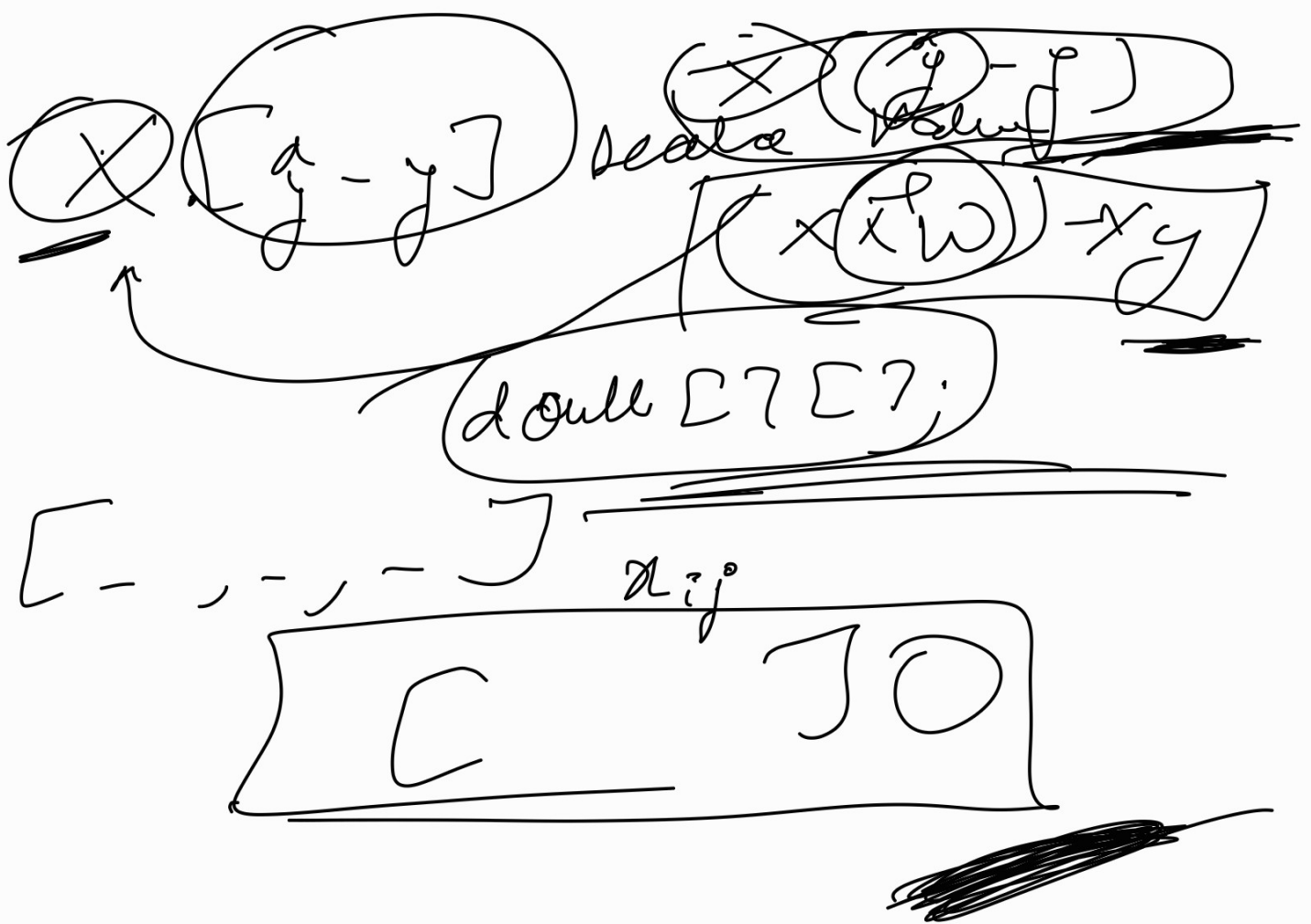
$$= \frac{1}{2n} \|X^T w - y\|^2$$

$$\begin{pmatrix} 0.1 \\ 0.2 \\ 0.3 \end{pmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \quad X \hat{y} - X y$$

$$\begin{bmatrix} 0.1 & 0.2 & 0.3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 0.6 \\ 0.6 \\ 0.6 \end{bmatrix}$$

W



$$w^{t+1} = w^t + \eta \nabla L$$

$$\Delta L$$

$$BCE = -\frac{1}{n} \left[\sum y_i (\log p_i) + \sum (1-y_i) \log(1-p_i) \right]$$

$$\frac{\partial L}{\partial p} = -\frac{1}{n} \left[\frac{y_i}{p_i} + \frac{(1-y_i)}{(1-p_i)} (-1) \right]$$

$$= -\frac{1}{n} \left[\frac{y_i}{p_i} - \frac{(1-y_i)}{(1-p_i)} \right]$$

$$= -\frac{1}{n} \left[\frac{y_i - \cancel{y_i p_i} - p_i + \cancel{y_i p_i}}{p_i (1-p_i)} \right]$$

$$= -\frac{1}{n} \left[\frac{y_i - p_i}{p_i (1-p_i)} \right]$$

$$= \frac{1}{n} \left[\frac{p_i - y_i}{p_i (1-p_i)} \right]$$

$$-\frac{1}{N} \left[y_i \log(p_i) + y_j \log(p_j) + y_k \log(p_k) \right]$$

$$-\frac{1}{N} \left[\frac{y_i}{p_i} + \frac{y_j}{p_j} + \frac{y_k}{p_k} \right]$$

$$-\frac{1}{N} \left[\frac{y_i (p_j \times p_k) + y_j (p_i \times p_k) + y_k (p_i \times p_j)}{p_i p_j p_k} \right]$$



$$\frac{e^{z_{ij}}}{\sum_{k=1}^C e^{z_{ik}}}$$

$$e^{z_{ij}} ()$$

$$\frac{e^{z_{ij}}}{e^{z_{ij}} + e^{z_{ij}} + e^{z_{ij}}} + () e^{z_{ij}}$$

$$2e$$

$$\frac{2 e^{z_{ij}} (e^{z_{ij}} + e^{z_{ij}})}{(e^{z_{ij}} + e^{z_{ij}} + e^{z_{ij}})}$$

Jacobian:

$$L = \frac{e^{z_i}}{\sum_{i=1}^C e^{z_i}}$$

where $C = \text{class}$

Let's say we have $C = 3$

$$L_i = \frac{e^{z_i}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

$$L_1 = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

$$\frac{\partial L_1}{\partial z_1} = \frac{e^{z_1} (e^{z_1} + e^{z_2} + e^{z_3}) - e^{z_1} \times e^{z_1}}{(e^{z_1} + e^{z_2} + e^{z_3})^2}$$

$$\frac{\partial L}{\partial z_1} = \frac{\cancel{e^{2z_1}} + e^{2z_1}(e^{z_1} + e^{z_3}) - \cancel{e^{2z_1}}}{(e^{z_1} + e^{z_2} + e^{z_3})^2}$$

$$= \frac{e^{2z_1}(e^{z_2} + e^{z_3})}{(e^{z_1} + e^{z_2} + e^{z_3})^2}$$

$$= \frac{e^{2z_1}}{e^{z_1} + e^{z_2} + e^{z_3}} \times \frac{e^{z_2} + e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

$$= y_1 \left[\frac{e^{z_1} + e^{z_2} + e^{z_3} - e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}} \right]$$

$$= y_1 \left[\frac{e^{z_1} + e^{z_2} + e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}} - \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}} \right]$$

$$= y_1 [1 - y_1]$$

$$\frac{\partial y_1}{\partial z_2} = \frac{0 - e^{z_1} (e^{z_2})}{(e^{z_1} + e^{z_2} + e^{z_3})^2}$$

$$= - \frac{e^{z_1} e^{z_2}}{(e^{z_1} + e^{z_2} + e^{z_3})^2}$$

$$= - \left[\frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}} \right] \left[\frac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3}} \right]$$

$$= -y_1 y_2$$

$$J = \begin{bmatrix} y_1(1-y_1) & -y_1 y_2 & -y_1 y_3 \\ -y_1 y_2 & y_2(1-y_2) & -y_2 y_3 \\ -y_1 y_3 & -y_2 y_3 & y_3(1-y_3) \end{bmatrix}$$

Jacobian Matrix of Softmax

Derivative for Coding:-

$$J = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \begin{bmatrix} 1-y_1 & -y_2 & -y_3 \\ -y_1 & 1-y_2 & -y_3 \\ -y_1 & -y_2 & 1-y_3 \end{bmatrix}$$

Hadamard Multiplication
OR

Element Wise Multiplication

$$J = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix}$$

$$J = \text{softmax}(z) * (\mathbf{I} - \text{softmax}(z))^T$$

$$\begin{bmatrix} | & | & | & | \\ | & | & | & | \\ | & | & | & | \end{bmatrix} = \begin{bmatrix} | \\ | \\ | \end{bmatrix}$$

3×4

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}$$

.Nadabard

4×3

(3×4)

$$\begin{bmatrix} -y_1 & -y_2 & -y_3 \end{bmatrix}$$

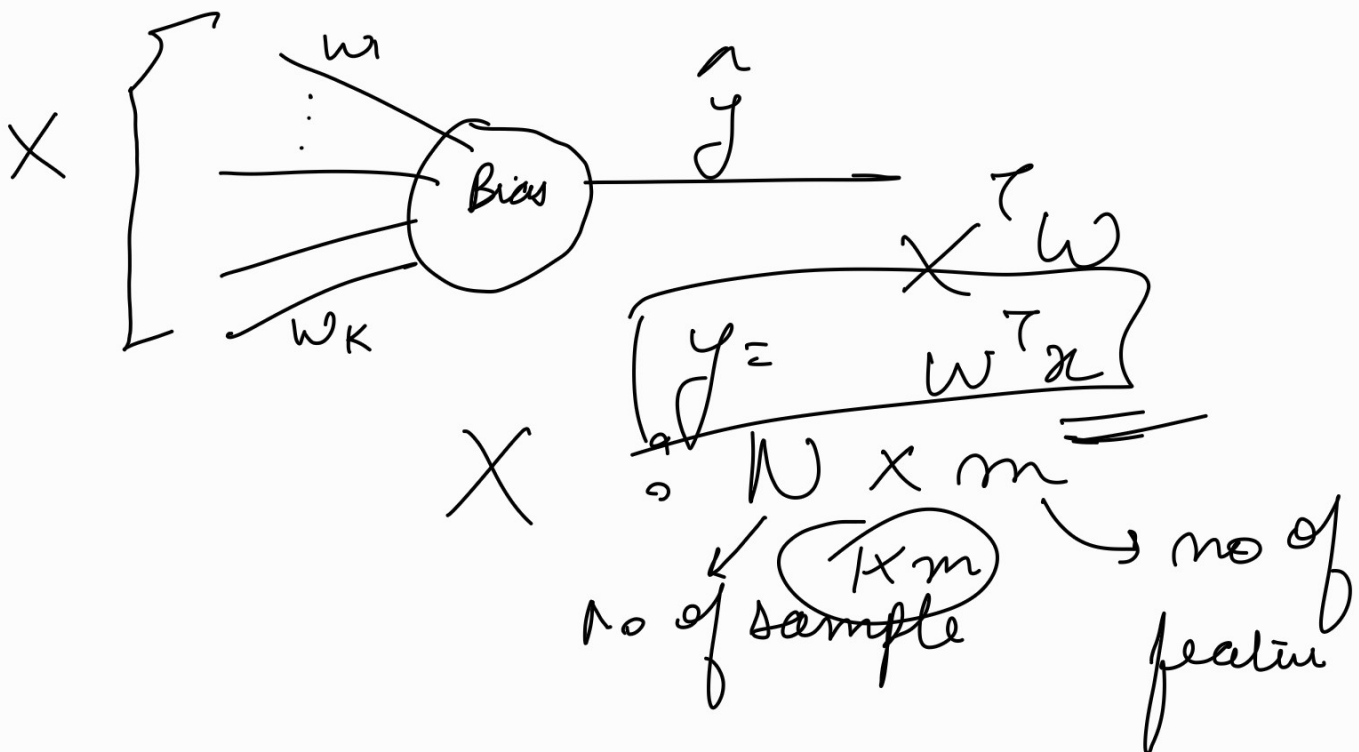
(4×4)

Perceptron

$$\hat{y} = \begin{cases} 1, & w^T x \geq 0 \\ -1, & w^T x < 0 \end{cases}$$

A lgo :- $\begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} \end{bmatrix} = \text{val}$

$$w^{T+1} = w^T + x_i y_i \quad y = \begin{cases} \end{cases}$$



$$w : 1 \times m$$

$$w^T X = m \times 1 \cdot 1 \times m$$

$$= m \times m$$

$$X = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ -2 & -1 \\ -1 & -2 \end{bmatrix} \quad \boxed{1 \times 1} \quad y = w^T x \quad w = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \end{bmatrix}_{1 \times 2} \begin{bmatrix} 2 & 1 & -2 & -1 \\ 1 & 2 & -1 & -2 \end{bmatrix}_{2 \times 4}$$

(w^T)

$$\begin{bmatrix} 0 \end{bmatrix}_{1 \times 1}$$

$$y = w^T x$$

$$w = 1 \times m$$

$$y = 1 \times 1$$

$$1 \times m \times m \times 1$$

$$\boxed{1 \times 1}$$

$$\begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix}$$

$$\boxed{X w^T} \rightarrow \boxed{1 \times m} \rightarrow \boxed{m \times 1} \quad \boxed{X w^T} \rightarrow \boxed{1 \times 1}$$

$$\boxed{w^{t+1} = w^t + \alpha y_i}$$

$$b^{t+1} = b^t + y_i$$

$$z = w^T x + b$$

$$\sigma(z) = \frac{1}{1 + e^{-z}} = p(z)$$

$$L = \prod_{i=1}^N (p(z_i))^{y_i} (1 - p(z_i))^{1 - y_i}$$

likelihood

$$\begin{aligned} \log L &= \sum_{i=1}^N y_i \log(p(z_i)) + \\ &\quad (1 - y_i) \log(1 - p(z_i)) \\ &= \sum_{i=1}^N y_i \log(p(z_i)) + \log(1 - p(z_i)) \\ &\quad - y_i \log(1 - p(z_i)) \end{aligned}$$

$$= \sum_{i=1}^n y_i \log \left(\frac{1}{1+e^{-z_i}} \right) + \log \left(1 - \frac{1}{1+e^{-z_i}} \right) \\ - y_i \log \left(1 - \frac{1}{1+e^{-z_i}} \right)$$

$$= \sum_{i=1}^n -y_i \log(1+e^{-z_i}) + \log \left(\frac{e^{-z_i}}{1+e^{-z_i}} \right) \\ - y_i \log \left(\frac{e^{-z_i}}{1+e^{-z_i}} \right)$$

$$= \sum_{i=1}^n \cancel{-y_i \log(1+e^{-z_i})} - z_i - \log(1+e^{-z_i}) \\ + y_i z_i + \cancel{y_i \log(1+e^{-z_i})}$$

$$= \sum_{i=1}^n -z_i - \log(1+e^{-z_i}) + y_i z_i$$

$$\frac{\partial L}{\partial z_i} = \sum_{i=1}^n -1 - \frac{1}{1+e^{-z_i}} (-e^{-z_i}) + y_i$$

$$= \sum_{i=1}^n -1 + \frac{e^{-z_i}}{1+e^{-z_i}} + y_i$$

$$= \sum_{i=1}^n y_i - \frac{1}{1+e^{-z_i}}$$

$$\nabla L = \sum_{i=1}^n y_i - p(z_i)$$

Linear :-

$$\boxed{w^T a + b} \rightarrow \boxed{1 \times 1}$$

$$\boxed{w^T a + b} \rightarrow \text{class} \times \text{class}$$

No of $\times$ Class
Sample

1 x output

$$x w^T$$

$$N \times C \times C \times D \quad \downarrow \quad 2$$

$$\boxed{N \times O} + \boxed{1 \times O}$$

$$n = 3$$

$$O = 2$$

$$X \rightarrow 2 \times 3$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix} + [1 \ 2]$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 5 \end{bmatrix}_{3 \times 2} + \begin{bmatrix} 1 & 2 \end{bmatrix}_{1 \times 2}$$

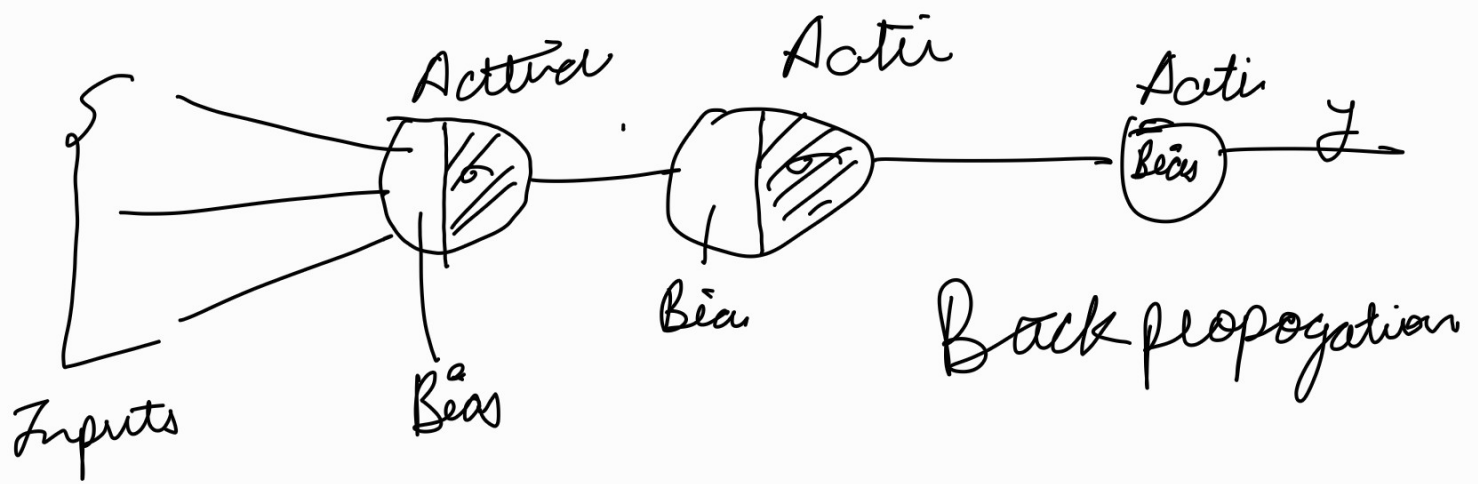
$$\begin{array}{c} i \quad j \longrightarrow \\ \downarrow \\ \begin{bmatrix} 14 & 23 \\ 23 & 38 \end{bmatrix}_{2 \times 2} + \begin{bmatrix} 1 & 2 \end{bmatrix} \end{array} \quad \begin{array}{l} 2+6+15 \\ 4+9=13+25 \\ \textcircled{38} \end{array}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ -1 & 2 \end{bmatrix}_{2 \times 3} \quad \begin{array}{l} 3+8 \quad 39 \\ 15+21 \quad 5+12 \end{array}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ -1 & 2 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 2 & 3 & 5 \\ 2 & 1 & 6 \end{bmatrix}_{2 \times 3} = \begin{bmatrix} 5 & 11 & 17 \\ 11 & 25 & 39 \\ 3 & 5 & 7 \end{bmatrix}$$

$\begin{array}{cc} 1+16 & 1+4 \end{array}$

$$\begin{array}{l} -1+7 \\ -3+8 \end{array} \quad \begin{array}{l} -5+12 \end{array} \quad \begin{bmatrix} 6 & 13 & 20 \\ 12 & 27 & 42 \\ 4 & 7 & 10 \end{bmatrix}$$



$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \times \frac{\partial a}{\partial z} \times \frac{\partial z}{\partial w}$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial a} \times \frac{\partial a}{\partial z} \times \frac{\partial z}{\partial b}$$

$$\frac{\partial L}{\partial a} = \frac{\partial L}{\partial a} \times \frac{\partial a}{\partial z} \times \frac{\partial z}{\partial a}$$

Eg:- $\boxed{Z = w^T a + b}$

$$\frac{\partial Z}{\partial w} = a, \quad \frac{\partial Z}{\partial b} = 1, \quad \frac{\partial Z}{\partial a} = w$$

$$a = \sigma(z)$$

$$\frac{\partial a}{\partial z} = \sigma(z)(1 - \sigma(z))$$

$$L = -[y \log(a) + (1-y) \log(1-a)]$$

$$\frac{\partial L}{\partial a} = - \left[\frac{y}{a} + \frac{(1-y)(-1)}{(1-a)} \right]$$

$$= - \left[\frac{y}{a} - \frac{(1-y)}{(1-a)} \right]$$

$$= - \left[\frac{y(1-a) - a(1-y)}{a(1-a)} \right]$$

$$= - \left[\frac{y - \cancel{ya} - a + \cancel{ya}}{a(1-a)} \right]$$

$$= \left[\frac{a-y}{a(1-a)} \right]$$

$$\frac{\partial L}{\partial w} = \left[\frac{a-y}{\cancel{a(1-a)}} \right] \times \cancel{a(1-a)} \times x$$

$$\boxed{\frac{\partial L}{\partial w} = x[a-y]}$$

$$\frac{\partial L}{\partial b} = \left[\frac{a-y}{\cancel{a(1-a)}} \right] \times \cancel{(1-a)} \times 1$$

$$= [a-y]$$

$$\frac{\partial L}{\partial a} = \left[\frac{a - y}{a(1-a)} \right] a(1-a) \times w$$

$$= w[a - y]$$

$$\frac{\partial L_{ij}}{\partial w_{jk}} = w_{jk} [a_{ij} - y_{ij}]$$

$$X = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad w = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$b = \begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$Z = Xw^T + b$$

$$= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$3 \times 2$$

$$= \begin{bmatrix} 5 & 11 \\ 11 & 25 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 2 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 13 \\ 12 & 27 \end{bmatrix}$$

$$dZ = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$dW = dZ^T \omega$$

$$= \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 2 \\ 4 \end{bmatrix}$$

2+12

$$= \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix}$$

4+16

$$dB = \begin{bmatrix} 4 & 6 \end{bmatrix}$$

$$3 \times 12$$

$$dX = dZ W$$

$$= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{matrix} 2 \times 8 \\ 6 \times 16 \end{matrix}$$

$$= \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$$

$$\frac{\partial L}{\partial a} = \left[\frac{a - y}{a(1-a)} \right]$$